To count or to predict

Taking arms against a sea of statistics

Erdin Mujezinović & Ruben van de Vijver

Department of General Linguistics
Heinrich-Heine University Düsseldorf, Germany

hhu Heinrich Heine Universität Düsseldorf

29 mfm fəˈnɒlədʒi

25th May 2022

To count or to predict

Taking arms against a sea of statistics

To count or to predict

Taking arms against a sea of statistics

How do native speakers acquire their language specific phonological categories?

To count or to predict

Taking arms against a sea of statistics

How do native speakers acquire their language specific phonological categories?

- Certain phonetic dimensions are used to differentiate meaning

To count or to predict

Taking arms against a sea of statistics

How do native speakers acquire their language specific phonological categories?

- Certain phonetic dimensions are used to differentiate meaning
- The same phonological category can vary in its phonetic implementation cross-linguistically

To count or to predict

Taking arms against a sea of statistics

How do native speakers acquire their language specific phonological categories?

- Certain phonetic dimensions are used to differentiate meaning
- The same phonological category can vary in its phonetic implementation cross-linguistically
- Understanding the learning mechanisms = understanding how and why these differences arise

To count or to predict

Taking arms against a sea of statistics

Two conceptualizations:

To count or to predict

Taking arms against a sea of statistics

Two conceptualizations:

Distributional learning

Contiguity: Counting co-occurrences (Maye, Werker, & Gerken, 2002; Saffran, 2020; Saffran, Aslin, & Newport, 1996; Saffran & Kirkham, 2018).

To count or to predict

Taking arms against a sea of statistics

Two conceptualizations:

Distributional learning

Contiguity: Counting co-occurrences (Maye, Werker, & Gerken, 2002; Saffran, 2020; Saffran, Aslin, & Newport, 1996; Saffran & Kirkham, 2018).

Error-driven learning

Contingency: Using cues to predict outcomes (Hoppe et al., 2022; Nixon, 2020; Nixon & Tomaschek, 2021; Ramscar, Dye, & McCauley, 2013; Rescorla & Wagner, 1972).

Distributional learning

Maye, Werker, & Gerken (2002)

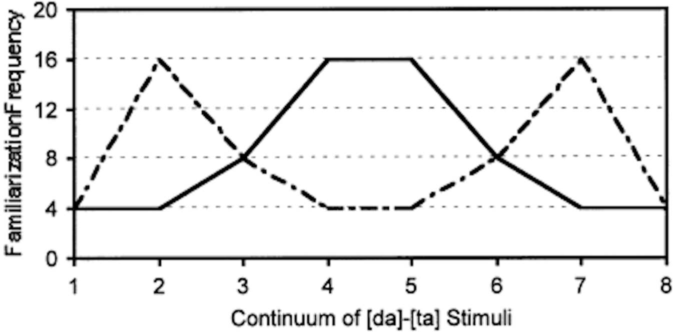

Figure 1: Unimodal (drawn line) vs. bimodal (dashed line) distribution (Maye, Werker, & Gerken, 2002)

Distributional learning

Maye, Werker, & Gerken (2002)

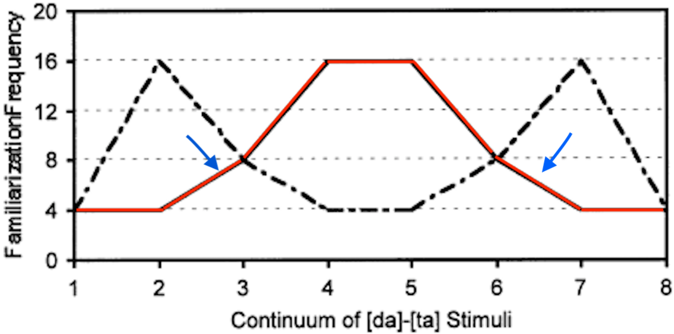

Figure 2: Unimodal (drawn line) vs. bimodal (dashed line) distribution (Maye, Werker, & Gerken, 2002)

- What if the tails are not symmetrical (not normally distributed?)

Error-driven distributional learning

Olejarczuk, Kapatsinski, & Baayen (2018)

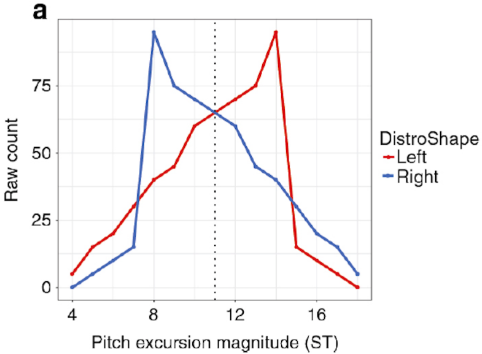

Figure 3: Distribution of LHL syllables ([ka]) for both groups in Olejarczuk, Kapatsinski, & Baayen (2018)

Error-driven distributional learning

Olejarczuk, Kapatsinski, & Baayen (2018)

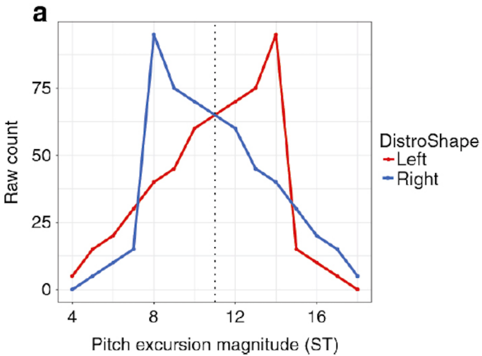

Figure 3: Distribution of LHL syllables ([ka]) for both groups in Olejarczuk, Kapatsinski, & Baayen (2018)

- Left = skew is in the lower excursions
- Right = skew is in the higher excursions

Error-driven distributional learning

Olejarczuk, Kapatsinski, & Baayen (2018)

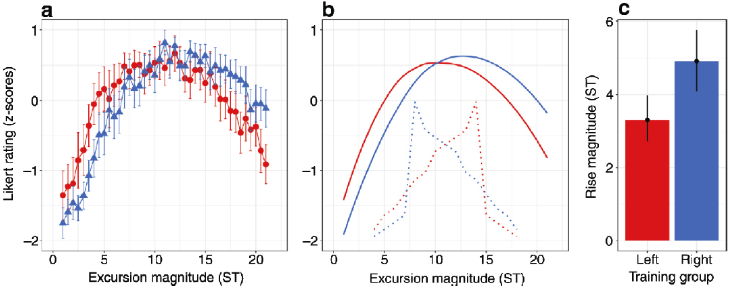

Figure 4: Results in Olejarczuk, Kapatsinski, & Baayen (2018)

Error-driven distributional learning

Olejarczuk, Kapatsinski, & Baayen (2018)

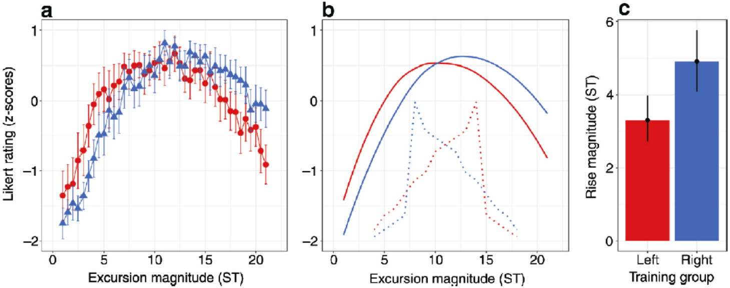

Figure 4: Results in Olejarczuk, Kapatsinski, & Baayen (2018)

Ratings/productions were shifted towards the tails: Surprisal has a stronger effect on the category

Our study

The goal

Replicating the results of Olejarczuk, Kapatsinski, & Baayen (2018) with...

Our study
The goal

Replicating the results of Olejarczuk, Kapatsinski, & Baayen (2018) with...

- a different learner group (German native speakers)

Our study
The goal

Replicating the results of Olejarczuk, Kapatsinski, & Baayen (2018) with...

- a different learner group (German native speakers)
- a novel phonetic dimension (consonant-gemination)

Our study

The goal

Replicating the results of Olejarczuk, Kapatsinski, & Baayen (2018) with...

- a different learner group (German native speakers)
- a novel phonetic dimension (consonant-gemination)
    - Original effect found for the learning of a rise-fall tonal category with English speakers

Our study
The goal

Replicating the results of Olejarczuk, Kapatsinski, & Baayen (2018) with...

- a different learner group (German native speakers)
- a novel phonetic dimension (consonant-gemination)
  - Original effect found for the learning of a rise-fall tonal category with English speakers
  - Problem: Pitch is not used at the lexical stratum in English

Our study
The goal

Replicating the results of Olejarczuk, Kapatsinski, & Baayen (2018) with...

- a different learner group (German native speakers)
- a novel phonetic dimension (consonant-gemination)
  - Original effect found for the learning of a rise-fall tonal category with English speakers
  - Problem: Pitch is not used at the lexical stratum in English
  - Effect found in Olejarczuk, Kapatsinski, & Baayen (2018) may be limited to non-lexical categories

Our study
The goal

Replicating the results of Olejarczuk, Kapatsinski, & Baayen (2018) with...

- a different learner group (German native speakers)
- a novel phonetic dimension (consonant-gemination)
  - Original effect found for the learning of a rise-fall tonal category with English speakers
  - Problem: Pitch is not used at the lexical stratum in English
  - Effect found in Olejarczuk, Kapatsinski, & Baayen (2018) may be limited to non-lexical categories (such as tone)

Our study

The goal

Replicating the results of Olejarczuk, Kapatsinski, & Baayen (2018) with...

- a different learner group (German native speakers)
- a novel phonetic dimension (consonant-gemination)
  - Original effect found for the learning of a rise-fall tonal category with English speakers
  - Problem: Pitch is not used at the lexical stratum in English
  - Effect found in Olejarczuk, Kapatsinski, & Baayen (2018) may be limited to non-lexical categories (such as tone)
  - German uses duration lexically for vowels (but not consonants)

Our study

The goal

Replicating the results of Olejarczuk, Kapatsinski, & Baayen (2018) with...

- a different learner group (German native speakers)
- a novel phonetic dimension (consonant-gemination)
  - Original effect found for the learning of a rise-fall tonal category with English speakers
  - Problem: Pitch is not used at the lexical stratum in English
  - Effect found in Olejarczuk, Kapatsinski, & Baayen (2018) may be limited to non-lexical categories (such as tone)
  - German uses duration lexically for vowels (but not consonants)
  - /a/ vs. /aː/ in *Bahn* [baːn] vs. *Bann* [ban]

Our study

What phonological category had to be learned?

Gemination

Segment duration of lateral approximant [lː] in [alːa]-sequence manipulated in 20 ms increments using **Praat** (Boersma & Weenink, 2021)

Our study

Research question

Research question

Do less frequent tokens have a stronger influence on the phonological category representation (in this case /ɪː/)?

Our study

Hypotheses

Three hypotheses:

Our study

Hypotheses

Three hypotheses:

1. Whole distribution **(the mean)** has the strongest effect

Our study

Hypotheses

Three hypotheses:

1. Whole distribution **(the mean)** has the strongest effect
   - **No difference** between groups

Our study
Hypotheses

Three hypotheses:

1. Whole distribution **(the mean)** has the strongest effect
   - **No difference** between groups
2. Most frequent tokens **(the modal values)** have the strongest effect

Our study

Hypotheses

Three hypotheses:

1. Whole distribution **(the mean)** has the strongest effect
   - **No difference** between groups
2. Most frequent tokens **(the modal values)** have the strongest effect
   - For Left skew = **higher durations** are more representative

Our study

Hypotheses

Three hypotheses:

1. Whole distribution **(the mean)** has the strongest effect
   - **No difference** between groups
2. Most frequent tokens **(the modal values)** have the strongest effect
   - For Left skew = **higher durations** are more representative
3. Rarer tokens **(the skew)** have the strongest effect

Our study

Hypotheses

Three hypotheses:

1. Whole distribution **(the mean)** has the strongest effect
   - **No difference** between groups
2. Most frequent tokens **(the modal values)** have the strongest effect
   - For Left skew = **higher durations** are more representative
3. Rarer tokens **(the skew)** have the strongest effect
   - For Left skew = **lower durations** are more representative

Our study
Hypotheses

Three hypotheses:

1. Whole distribution **(the mean)** has the strongest effect
   - **No difference** between groups
2. Most frequent tokens **(the modal values)** have the strongest effect
   - For Left skew = **higher durations** are more representative
3. Rarer tokens **(the skew)** have the strongest effect
   - For Left skew = **lower durations** are more representative

The former two are in line with distributional learning (count), the latter one with error-driven learning (predict)

Our study

The stimuli

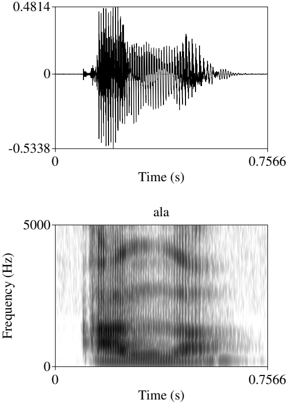

Figure 5: The original [alːa]-stimulus

Our study

The stimuli

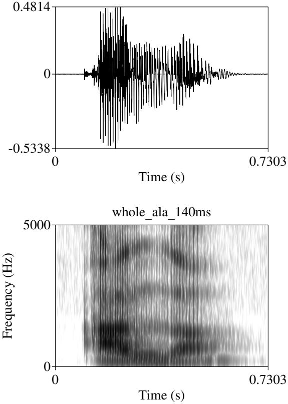

Figure 6: Shorter [alːa]-stimulus (140 ms)

Our study

The stimuli

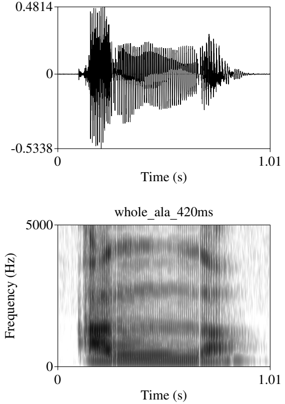

Figure 7: Longer [alːa]-stimulus (420 ms)

Our study

The experimental groups

Group	Range	Modal	Mean	Tokens
Right-skew	160-420 ms	220 ms	280 ms	256
Left-skew	140-400 ms	340 ms	280 ms	256
Test	80-480 ms			21

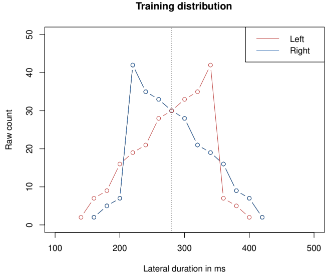

Figure 8: Distribution of the different segment durations

Our study

The experimental groups

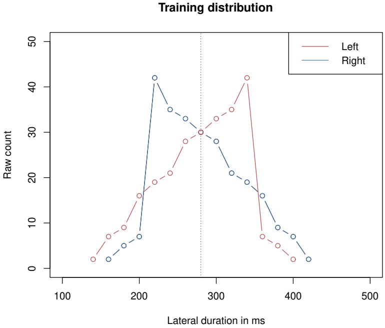

Figure 9: Right skew (blue) and left skew (red)

Our study

The experimental groups

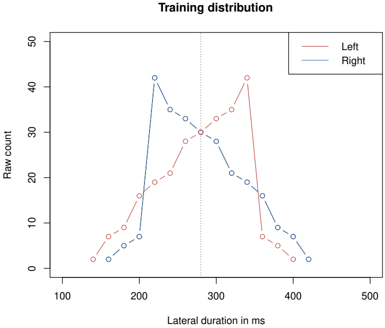

Figure 9: Right skew (blue) and left skew (red)

1. Participants in both groups had to judge the [alːa] tokens for their *Category Goodness* using a 7-point Likert-scale

Our study

The experimental groups

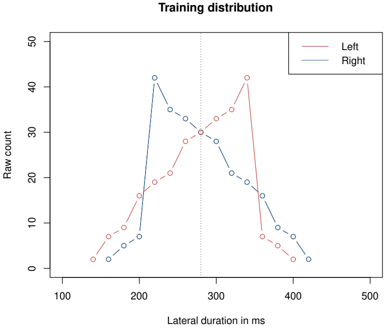

Figure 9: Right skew (blue) and left skew (red)

1. Participants in both groups had to judge the [alːa] tokens for their *Category Goodness* using a 7-point Likert-scale
2. Participants had to produce three representative [alːa]-tokens

Our study

Results I: Category-Goodness

Overall: 294 data points (14 participants – 9x Left-skew, 5x Right-skew)

Our study

Results I: Category-Goodness

Overall: 294 data points (14 participants – 9x Left-skew, 5x Right-skew)

- Left-skew: 189 observations (9 participants x 21 stimuli)

Our study

Results I: Category-Goodness

Overall: 294 data points (14 participants – 9x Left-skew, 5x Right-skew)

- Left-skew: 189 observations (9 participants x 21 stimuli)
- Right-skew: 105 observations (5 participants x 21 Stimuli)

Our study

Results I: Category-Goodness

Overall: 294 data points (14 participants – 9x Left-skew, 5x Right-skew)

- Left-skew: 189 observations (9 participants x 21 stimuli)
- Right-skew: 105 observations (5 participants x 21 Stimuli)
- Dataanalysis done in R (R Core Team, 2021)

Our study

Results I: Category-Goodness

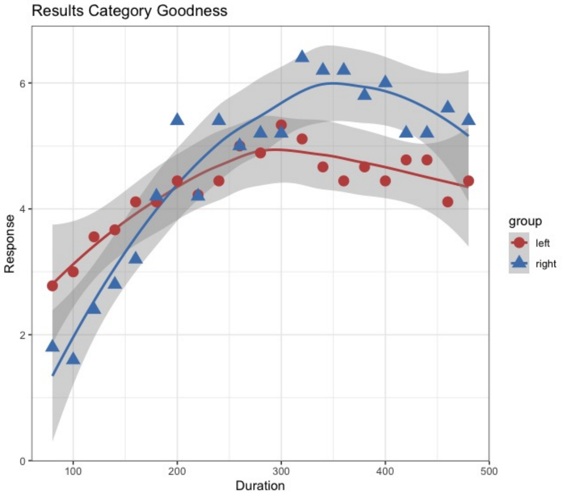

Figure 10: Overall results 7-point Likert scale (grey = Standard Error)

Our study

Results I: Category-Goodness

Table 1: Summary table GAM Model (mgcv-package (Wood, 2011)).

A.Parametric coefficients					
	Estimate	Std. Error	t-value	Pr(>|t|)	
(Intercept)	4.1846	0.2814	14.87	< 0.0001	***
B.Smooth terms					
	edf	Ref.df	F	p-value	
s(stimulus):groupleft	1.996	2.230	2.549	0.08985	.
s(stimulus):groupright	2.161	2.429	4.816	0.00762	**
s(stimulus,subject)	62.890	125.000	3.020	< 0.0001	***

Our study

Results II: Production

Overall: 42 data points (14 participants – 9x Left-skew, 5x Right-skew)

Our study

Results II: Production

Overall: 42 data points (14 participants – 9x Left-skew, 5x Right-skew)

- Left-skew: 27 observations (9 participants x 3 stimuli)

Our study

Results II: Production

Overall: 42 data points (14 participants – 9x Left-skew, 5x Right-skew)

- Left-skew: 27 observations (9 participants x 3 stimuli)
- Right-skew: 15 observations (5 participants x 3 stimuli)

Our study

Results II: Production

Overall: 42 data points (14 participants – 9x Left-skew, 5x Right-skew)

- Left-skew: 27 observations (9 participants x 3 stimuli)
- Right-skew: 15 observations (5 participants x 3 stimuli)
- Dataanalysis in R (R Core Team, 2021)

Our study

Results II: Production

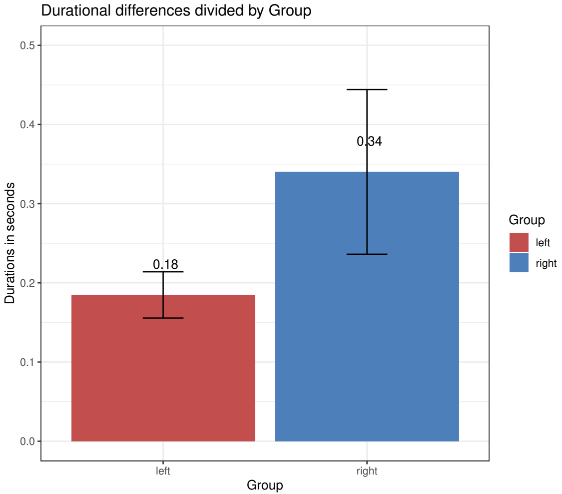

Figure 11: Difference in consonant duration between groups

Our study

Results II: Production

Linear mixed effects model computed using `lmerTest`-package (Kuznetsova, Brockhoff, & Christensen, 2017):

Our study

Results II: Production

Linear mixed effects model computed using `lmerTest`-package (Kuznetsova, Brockhoff, & Christensen, 2017):

- Random intercepts for subject and item

Our study

Results II: Production

Linear mixed effects model computed using `lmerTest`-package (Kuznetsova, Brockhoff, & Christensen, 2017):

- Random intercepts for subject and item
- Significant effect for group (Est. = 0.1575, SE = 0.0630, df = 12.0041, = t = 2.501, p = 0.02785)

Our study

Results II: Production

Linear mixed effects model computed using `lmerTest`-package (Kuznetsova, Brockhoff, & Christensen, 2017):

- Random intercepts for subject and item
- Significant effect for group (Est. = 0.1575, SE = 0.0630, df = 12.0041, = t = 2.501, p = 0.02785)
- Consonant duration differed between groups!

Discussion

Comparison to Olejarczuk, Kapatsinski, & Baayen (2018)

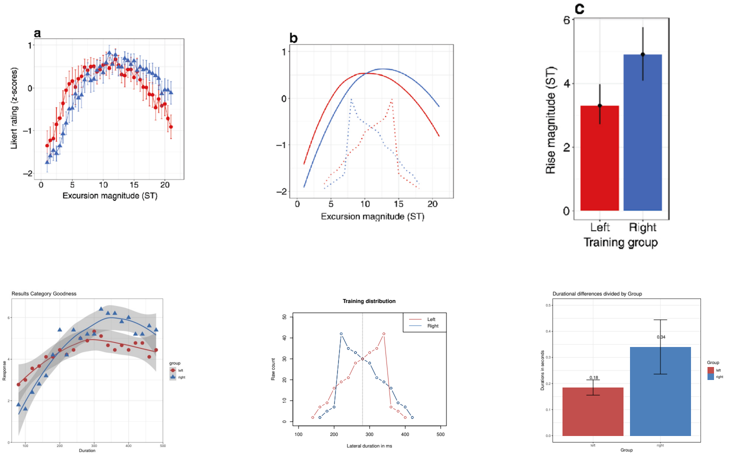

Figure 12: Comparison original study vs. replication

Conclusion

Successful replication of Olejarczuk, Kapatsinski, & Baayen (2018) with consonant-gemination!

Conclusion

Successful replication of Olejarczuk, Kapatsinski, & Baayen (2018) with consonant-gemination!

- Less frequent tokens have a stronger influence on the phonological category ✓

Conclusion

Successful replication of Olejarczuk, Kapatsinski, & Baayen (2018) with consonant-gemination!

- Less frequent tokens have a stronger influence on the phonological category ✓
- Learners do not simply count co-occurrences, but they instead **predict** subsequent tokens (outcomes) by using previous tokens as cues
    - Error-driven learning predicts such an effect

Conclusion

Successful replication of Olejarczuk, Kapatsinski, & Baayen (2018) with consonant-gemination!

- Less frequent tokens have a stronger influence on the phonological category ✓
- Learners do not simply count co-occurrences, but they instead **predict** subsequent tokens (outcomes) by using previous tokens as cues
  - Error-driven learning predicts such an effect
  - Rare category tokens are more surprising, generate more error and therefore cause stronger adjustment of the phonological category

Conclusion

Successful replication of Olejarczuk, Kapatsinski, & Baayen (2018) with consonant-gemination!

- Less frequent tokens have a stronger influence on the phonological category ✓
- Learners do not simply count co-occurrences, but they instead **predict** subsequent tokens (outcomes) by using previous tokens as cues
  - Error-driven learning predicts such an effect
  - Rare category tokens are more surprising, generate more error and therefore cause stronger adjustment of the phonological category
  - Frequent tokens define the category, but rarer instances constrain it!

Conclusion

Thank you for your attention!

References I

- Boersma, P., & Weenink, D. (2021). *Praat: Doing phonetics by computer [Computer program]*. Version 6.1.50.
- Hoppe, D. B., Hendriks, P., Ramscar, M., & van Rij, J. (2022). An exploration of error-driven learning in simple two-layer networks from a discriminative learning perspective. *Behavior Research Methods*, 1–31.
- Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. B. (2017). lmerTest package: Tests in linear mixed effects models. *Journal of Statistical Software*, *82*(13), 1–26.
- Maye, J., Werker, J. F., & Gerken, L. (2002). Infant sensitivity to distributional information can affect phonetic discrimination. *Cognition*, *82*(3), B101–B111.
- Nixon, J. S. (2020). Of mice and men: Speech sound acquisition as discriminative learning from prediction error, not just statistical tracking. *Cognition*, *197*, 104081.
- Nixon, J. S., & Tomaschek, F. (2021). Prediction and error in early infant speech learning: A speech acquisition model. *Cognition*, *212*, 104697.

References II

- Olejarczuk, P., Kapatsinski, V., & Baayen, R. H. (2018). Distributional learning is error-driven: The role of surprise in the acquisition of phonetic categories. *Linguistics Vanguard*, *4*(s2).
- R Core Team. (2021). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna, Austria.
- Ramscar, M., Dye, M., & McCauley, S. M. (2013). Error and expectation in language learning: The curious absence of mouses in adult speech. *Language*, 760–793.
- Rescorla, R. A., & Wagner, A. R. (1972). A theory of pavlovian conditioning: Variations in the effectiveness of reinforcement and nonreinforcement (A. H. Black & W. Prokasy, Eds.). *Classical conditioning II: current research and theory*, 64–99.
- Saffran, J. R. (2020). Statistical language learning in infancy. *Child Development Perspectives*, *14*(1), 49–54.
- Saffran, J. R., Aslin, R. N., & Newport, E. L. (1996). Statistical learning by 8-month-old infants. *Science*, *274*(5294), 1926–1928.

References III

Saffran, J. R., & Kirkham, N. Z. (2018). Infant statistical learning. *Annual review of psychology*, *69*, 181–203.

Wood, S. N. (2011). Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models. *Journal of the Royal Statistical Society (B)*, *73*(1), 3–36.

Conclusion

Successful replication of Olejarczuk, Kapatsinski, & Baayen (2018) with consonant-gemination!

- Less frequent tokens have a stronger influence on the phonetic category ✓
- Learners do not simply count co-occurrences, but they instead **predict** subsequent tokens (outcomes) by using previous tokens as cues
  - Such effects are naturally predicted by **Error-driven learning**
  - Rare category tokens are more surprising, generate more error and therefore cause stronger adjustment of the phonological category
- Frequent tokens define the category, but rarer instances constrain it!

Appendix

Results I: Category-Goodness

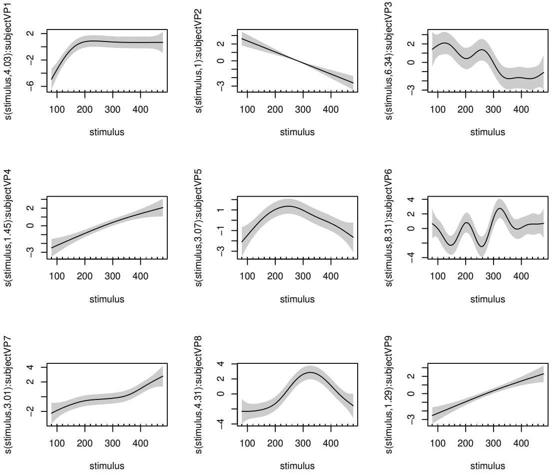

Figure 13: Individual results 7-point Likert scale (Left-skew)

Appendix

Results I: Category-Goodness

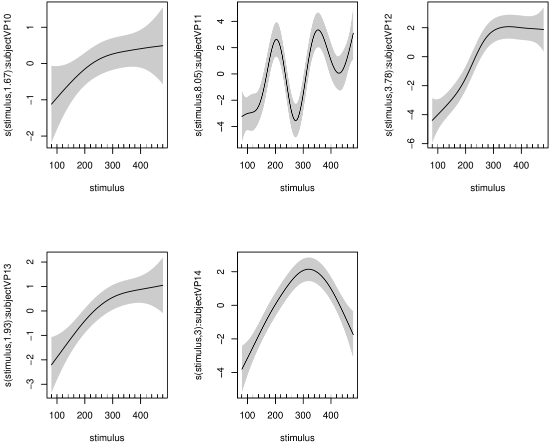

Figure 14: Individual results 7-point Likert scale (Right-skew)

Appendix

Results II: Production

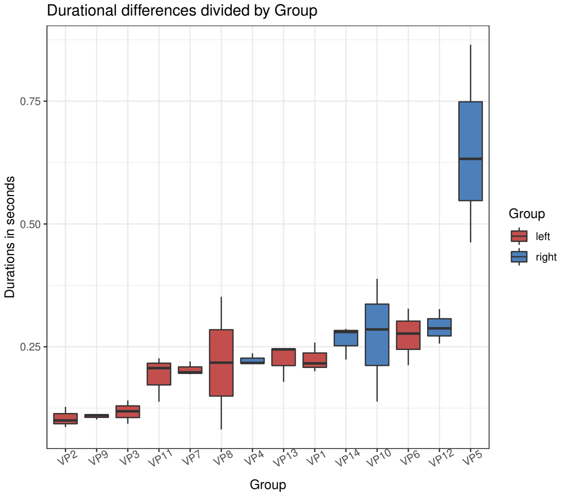

Figure 15: Difference in consonant duration between participants